

Checkpoint

13.1: Vector Functions

Problem 1. If $\vec{r}(t) = \langle t, e^t, e^{-t} \rangle$, what is $\vec{r}'(0)$ and what does it mean?

Problem 2. what is $\lim_{t \rightarrow 3} \langle \frac{1}{t} + \frac{1}{t^2} + \frac{1}{t^3} \rangle$

Problem 3. If $\vec{r}'(t) = (\sqrt{t})\hat{i} + (\sqrt{2t})\hat{j} + (\sqrt{3t})\hat{k}$, what is the corresponding unit tangent vector?

Problem 4. Find $\vec{r}''(t)$ where $\vec{r}(t) = \langle \ln t, \sin t, \tan t \rangle$.

Problem 5. Find $\int \vec{r}(t)dt$ where $\vec{r}(t) = \langle e^{3t}, t^2, \cos t \rangle$

1 Answers

1. $\vec{r}(0) = \langle 0, 1, 1 \rangle$ it is a 3-dimensional position vector
2. $\langle 1/3, 1/9, 1/27 \rangle$
3. $T(t) = \frac{1}{\sqrt{6t}}((\sqrt{t})\hat{i} + (\sqrt{2t})\hat{j} + (\sqrt{3t})\hat{k})$
4. $\vec{r}'(t) = \langle \frac{1}{t}, \cos t, \sec^2 t \rangle$
5. $\int \vec{r}(t)dt = \langle \frac{1}{3}e^{3t}, \frac{1}{3}t^3, \sin t \rangle + \langle c_1, c_2, c_3 \rangle$ or $+C$ where C is a vector.